AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM105
Introduction to Computer Programming

Final Examination
Fall Semester 2025/2026

Date: 12 / Jan. / 2026

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **100**
- This exam consists of **5** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

Question 1 [10 Marks]

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For each question, choose the most appropriate answer. Copy the answer to your answer booklet. Each question carries 2 marks.

1. Which operation will be executed last in the following expression:
```
double x = (Math.sqrt( 4 ) + 7) / (2 * 10 - 5);
```
a. /
b. +
c. -
d. *

2. Which of the following is not a Java reserved word?
a. break
b. continue
c. args
d. void

3. In the below code, which line has error?
```
1 for (int i=1; i!= 5; i++)
2   System.out.println(i);
3 System.out.println(i);
```
a. 1
b. 2
c. 3
d. There is no error

4. What is the value stored in `a[1]` after the executing the following Java statement?
```
int [] a  = new int [10];
```
a. 10
b. 1
c. 0
d. It is not initialized

5. The `do-while` loop in Java executes at least once, even if the condition is false.
a. True
b. False

Question 2 [25 Marks]

Trace the following pieces of code and write their exact output:

1. (4 Marks)
```
for (int i = 1; i <= 4; i++) {
  for (int j = 1; j <= i; j++)
    System.out.print(1);
  System.out.println();
}
```

2. (5 Marks)
```
int a = 10;
while (a>1){
   System.out.println(a);
   if (a<3)
      break;
   a=a-2;
}
```


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3. (5 Marks)

```
public class Exam {
    public static void main(String[] args) {
        int i = 1, j = 7;
        System.out.println(calculate());
        System.out.println(calculate(i));
        System.out.println(calculate(j));
        System.out.println(calculate(i,j));
        System.out.println(calculate(j,i));
    }
    public static int calculate() {
        return 0;
    }
    public static int calculate(int x) {
        x++;
        return x+x;
    }
    public static int calculate(int x,int y) {
        x/=y;
        return x+x;
    }
}
```

4. (5 Marks)

```
int j;
for (int i=1;i<10;i+=2){
  j =(i%3==0)?0:1;
  System.out.println(j);
}
```

5. (6 Marks)

```
int[] x = {10,3,4,8,5,1};
x[0]=x[1]++;
x[2]=x[3]/3;
x[3] = x[x[4]];
x[4]*=2;
x[5]--;
for (int i = 0; i <= 5; i++)
   System.out.println(x[i]);
```


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Question 3 [20 Marks]

Consider the following Java code that starts by declaring and initializing a 2D array `sales` representing the sales (in KD) made by 3 salespersons during 4 weeks. Complete the code to do the following:

1. Find the highest sales amount.
2. Calculate the average sales of the first week for the 3 salespersons rounded to 2 decimal places (i.e. the average of the sales in the first column).

```
int[][] sales = { {250, 300, 280, 320},
                  {400, 380, 390, 410},
                  {200, 250, 240, 260}
                };
int highest;

/* Write here the code to find the highest sales amount */
.....
.....

System.out.println("The highest sales amount is: " + highest);

int sum;
double average;

/* Write here the code to calculate and print the average sales
of the first week */
.....
.....
.....
.....
```

For questions 4 and 5 in the next page, copy the below code into your answer booklet and continue the rest of the code according to the given specifications.

```
import java.util.Scanner;
public class FinalExam {
      public static void main(String[] args) {
          Scanner input = new Scanner(System.in);

    }
}
```


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Question 4 [20 Marks]

Write Java code to do the following:

1. Ask the user to enter the number of words he/she wants to enter (n).
2. Read n words from the user.
3. For every word entered, count the number of A's in the words regardless of the case (uppercase and lowercase both count).
4. Print the result.

Here is a sample of the program run to help you understand how the program works:

```
Enter the number of words: 4
Enter word 1: Hello
Number of A's in "Hello": 0
Enter word 2: Anas
Number of A's in "Anas": 2
Enter word 3: Take
Number of A's in "Take": 1
Enter word 4: Banana
Number of A's in "Banana": 3
```

Question 5 [25 Marks]

Write Java code to do the following:

1. Write a Java method called `countPassing()` that takes an array of integer grades as an argument and returns the number of students who passed (grade ≥ 50).

2. In the main method, create a one-dimensional array of 6 integers.

3. Initialize the array with 6 grades entered by the user.

4. Call the method `countPassing()` and print the result.

Here are 2 samples of the program run to understand how the program works:

Program run 1:

```
Enter 6 grades: 90 59 75 44 61 25
Number of students who passed: 4
```

Program run 2:

```
Enter 6 grades: 33 45 66 79 80 20 12
Number of students who passed: 3
```

End of questions

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